Luca **Di Vita**

Profile

Machine Learning Engineer with a degree in Computer Engineering and a strong background in **Python** and **software engineering**. I design, build, and deploy **production-grade ML systems**, working across the entire lifecycle—from data pipelines to real-time inference and scalable microservices. Curious and research-driven, I also explore **evolutionary computation** and experimental approaches to AI. Active in the tech community as GDG Pescara organizer and Python Pescara co-founder, I frequently deliver technical talks and contribute to training and knowledge-sharing initiatives, aiming to make complex concepts accessible and practical.

Contacts

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Personal Information

Citizenship: **Italian**

Languages: **Italian** (native), **English** (professional working proficiency)

Skills

- **Machine Learning & Data**: data processing and feature engineering, model development and evaluation; scikit-learn, PyTorch, Keras, Pandas, NumPy.
- **Backend & MLOps**: Python, Flask, FastAPI, Docker; development of APIs and microservices for ML deployment.
- **Data Engineering**: SQL, data pipelines, ETL workflows.
- **Tools & Dev Practices**: Git, containerized environments, testing, CI-oriented workflows.
- **Soft skills**: effective communication with stakeholders, teamwork in cross-functional environments, structured problem solving.

Experience

Senior Machine Learning Engineer at *Cy4Gate (ELT Group)* — **2023.11–present**

◇ Initially joined as a consultant via Frontiere (2023.11–2024.08), then hired internally (2024.09–present). Development of **anomaly detection** pipelines for the RTA SIEM platform in **on-premise** environments, based on **scikit-learn** and **River**, integrated into low-latency **Docker** microservices with **Kafka** and **Elasticsearch**. Development of **RAG** and **agentic systems**, **MCP** servers, and **multimodal analysis** services for audio, video, images, and text, using models deployed entirely **on-premise**.

Lecturer (Freelance) at *ITS Lanciano* — **2024.04–2024.06**

◇ Teaching **object-oriented programming** in Python.

Senior Machine Learning Engineer (Freelance) at *20Tab* — **2023.08–present**

◇ Development of *CONNECT*, a semantic search system for scientific articles based on **Python**, **Hugging Face** models, and the **Qdrant** vector database, with attention-based explainability mechanisms.

◇ Development and fine-tuning of **YOLO** models for the CNR *AUTOMA* project, with **data augmentation** pipelines for the identification of invasive species in Italian waters.

Middle Machine Learning Engineer at *Frontiere* — **2023.03–2023.11**

◇ Development of containerized ML microservices using **Docker** for **computer vision** and **NLP** applications, including human detection and image classification with **YOLO/CNN**, and email classification based on **BERT embeddings**.

Junior Machine Learning Engineer at *Aesys* — **2019.11–2023.03**

◇ Development of ML solutions for enterprise clients in the insurance, retail, and industrial sectors, covering **NLP**, **OCR**, document classification, forecasting, and predictive maintenance, using **scikit-learn**, **Keras**, **SpaCy**, **SHAP**, and cloud services on **AWS/GCP**.

Research Fellow at *University of L'Aquila* — **2019.04–2019.11**

◇ Research on **Deep Reinforcement Learning** applied to Cyber Security through **Mininet** simulations and malware traffic analysis from `.pcap` files, using **Keras** and **Q-Learning**.

Education

Master's Degree in Computer Engineering (110/110) *University of L'Aquila.* — **2016–2018**

◇ Thesis: *Deep Reinforcement Learning applied to Cyber Security.*

◇ Main study areas: software engineering, algorithms and data structures, advanced databases, machine learning.

Publications Talks

Enabling Research Collaboration with AI: The BI4E Experience with Large Language Models *CEUR-WS* — **2025**

◇ Description of the CONNECT platform, based on transformer models and semantic embeddings for identifying potential academic collaborators from Scopus-indexed publications.

MTA-KDD'19: A Dataset for Malware Traffic Detection *CEUR-WS* — **2020**

◇ Updated dataset for malware traffic analysis, built through the collection, cleaning, and preprocessing of large volumes of network traffic for training machine learning models.

A Spherical Directional Anemometer Sensor System *MDPI* — **2017**

◇ Proposal and analysis of a compact directional anemometer without moving mechanical parts, based on differential pressure measurements using inductive transducers.

Technical articles, educational content, and talks available at: lucadivita.it/en/

Hobbies

- *Training and running:* gym workouts and running as part of my routine.
- *Reading, studying and blogging:* fiction and technical reading, continuous learning, and writing technical and educational articles.
- *Tech community:* active participant, event organizer, and speaker.
- *Gaming:* gaming in my free time.